Kausar Ahmed

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Computer Science graduate conducting academic research and AI development. Through research, hackathons, and hands-on projects, I have explored how AI can address practical problems. My primary research interest is NLP, particularly RAG and Information Retrieval. I enjoy developing AI systems that are efficient, reliable, and useful in real-world.

Education

University of Liberal Arts Bangladesh (ULAB) Oct. 2022 – Sep. 2026

BSc. in Computer Science and Engineering (CSE) | CGPA: 3.89 / 4.00

- Co-curricular: ULAB Computer Programming Club (UCPC)'s **Active Volunteer Award** recipient.

Publications

Peer-Reviewed Journal Articles:

[1] **Ahmed, K.**, Gomes, B. R., Hossain, S. M. J., Musa, E. K., Bhoyan, F. H., Mehedi, M. H. K., & Uddin, J. (2026). **Reducing False Alarms in Fire Detection Systems with YOLOv11 and Multi-Sensor Validation**. *IJEEE*, *22*(03), 3733. doi:10.22068/IJEEE.22.3.3733

Conference Proceedings:

[2] **Ahmed, K.**, Gomes, B. R., Hossain, S. M. J., Musa, E. K., Abdal, M. N., & Akhter, S. (2025). **ResNet50 for Glaucoma Detection: Hyperparameter-Tuned Transfer Learning on Retinal OCT Images**. In *16th ICCCNT*. (In Press).

Manuscripts Under Review:

[3] **Ahmed, K.**, Gomes, B. R., Hossain, S. M. J., & Quaderi, S. J. S. (2026). **MetricRouter: Retraining-Free Lightweight Adaptive Query Routing for Multi-Pipeline RAG** in ICCIT.

[4] **Ahmed, K.**, Choudhury, T., & Islam, M. K. (2026). **Effectiveness of Offline Data Augmentation in Cross-Domain Few-Shot Metric Learning** in ICCIT.

Manuscript in preparation:

[5] **Ahmed, K.**, Gomes, B. R., Hossain, S. M. J., Islam, T., Musa, E. K., & Quaderi, S. J. S. (2026). **D2RAG: A Context-Enriched Hierarchical RAG Architecture for Statutory Legal Corpora**.

Presentations

Talks & Panels:

[1] **Ahmed, K.** (2025, November). *LMS Standardization in Higher Education*. Invited student panelist at the Educational Innovation Symposium 2025, Center for Excellence in Teaching and Learning (CETL), ULAB, Dhaka, Bangladesh.

Posters:

[2] **Ahmed, K.**, Ahmed, N. U. (2025, November). *The Proof-of-Learning Framework: A Defense-Based Assessment Model for the AI Era*. Poster presented at the Educational Innovation Symposium 2025, Center for Excellence in Teaching and Learning (CETL), ULAB, Dhaka, Bangladesh.

Professional Experience

Research Assistant (RA) Sep. 2026 – Present

Dept. of CSE | University of Liberal Arts Bangladesh (ULAB)

- Conducting research with faculty members across ongoing projects in NLP and RAG.
- Mentoring undergraduate capstone project and thesis teams to shape research designs, refine experimental pipelines, and prepare manuscripts for publication.

Undergraduate Teaching Assistant (UTA) Jan. 2026 – Sep. 2026

Dept. of CSE | University of Liberal Arts Bangladesh (ULAB)

- Supported faculty members in conducting lab courses of approximately 40 students per class. Mentored students in programming fundamentals, debugged code, and reinforced concepts.

Research Experience

Assertion-Based Probing for Sensitive Information Disclosure in Black-Box RAG Systems Aug. 2026 – Present

Remote Research

In collaboration with Dr. Prasanthi Sreekumari | University of Louisiana at Monroe

Investigating whether false-premise queries can influence retrieval behavior or disclose protected information in black-box retrieval-augmented generation (RAG) systems. The research aims to develop a probing methodology to assess the robustness against adversarial queries.

MetricRouter: Retraining-Free Lightweight Adaptive Query Routing for Multi-Pipeline RAG May 2026 – Sep. 2026

Undergraduate Thesis | Supervisor: Shah Jafor Sadeek Quaderi

Investigated lightweight query routing mechanisms to mitigate the computational overhead of LLM classifiers. Proposed a metric-learning architecture coupled with a feedback-gated adaptation module to absorb corrections without retraining achieving a 0.9420 macro-F1.

LawB: An Interpretable RAG-Based Legal Assistant for Bangladesh's Citizens Jun. 2025 – May 2026

Undergraduate Capstone Project | Supervisor: Shah Jafor Sadeek Quaderi

Proposed **D2RAG**, a two-stage hierarchical retrieval architecture introduceing metadata-scoped filtering, enriching the corpus for better semantic matching across highly correlated legal texts. The methodology demonstrated a 7% improvement in response completeness over baseline.

Projects

Bilingual Multimodal RAG Learning Assistant for the NCTB Curriculum | GitHub May - Jun. 2026

Python, FAISS, BM25, ColPali, Gemini, Ollama, PyTorch

- Built a Bengali/English conversational RAG pipeline with LLM-based query rewriting to resolve context-dependent follow-up questions before retrieval.

LawB: Interpretable Hierarchical Legal Assistant Jun. 2025 – May 2026

Python, Flask, PyTorch, BGE-M3, Cross-Encoder, Gemma, Ollama, RAG

- Designed **D2RAG**, a dual-index hierarchical RAG architecture for statutory legal corpora with act-level index pruning followed by section-level retrieval and cross-encoder reranking.
- Developed **LawB**, an end-to-end legal assistant for Bangladeshi citizens providing verifiable statutory citations with reduced hallucination rates.

EXIREN: AIoT Fire Detection System | GitHub Jul. 2025 – Sep. 2025
Python, Flask, YOLOv11, OpenCV, PyTorch, IoT Sensors

- Developed an AIoT-based early fire detection system combining custom YOLOv11 computer vision with multi-sensor IoT validation.

Honors and Awards

Finalist | Infinity AI BuildFest 2026, BRAC University Jun. 2026

- Built a multimodal conversational RAG learning assistant for the NCTB curriculum (Top 9%).

Champion | ULAB Earth Hackathon 2026 Apr. 2026

- Developed an agricultural advisory integrating ML crop predictions with NASA Earthdata.

Finalist | Hult Prize 2025 at ULAB Dec. 2025

- Pitched an AI-assisted SaaS, designed for early veterinary consultation and pet health triage.

Honorable Mention (AI Category) & National Nominee | Blockchain Olympiad Bangladesh 2025 Sep. 2025

- Officially nominated to represent Bangladesh at the International Data Science Olympiad 2025 in Hong Kong.

1st Runner-up | ULAB CSE FEST Hackathon 2025 Jul. 2025

- Developed a full-stack sustainable recycling web platform in an intensive hackathon sprint.

Skills

Programming Languages: Python, Java, C, JavaScript

NLP & Information Retrieval: RAG, Hybrid Retrieval, LLMs (Gemma), BERT

Machine Learning & Frameworks: PyTorch, Scikit-learn, OpenCV, YOLO, ResNet

Web Development & Database: Flask, REST APIs, MySQL, Vercel

Tools, Platforms & Environments: Git, GitHub, LaTeX, Linux, Ollama

Licenses & Certifications

GitHub Foundations Certification, GitHub Sep. 2024

Community Service

Volunteer | International Youth Change Maker (IYCM) Oct. 2023 – Present
Youth Leadership and Community Development

- Supported youth-led community programs focused on children's rights, educational equity, and social development.